

Modelling Profit Sensitivity to Discounting in E-Commerce Transactions Using Linear and Non-Linear Machine Learning Approaches

Author:	Utsav Chand
Student ID:	25123856
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Abstract

Discounting has become a main tool in e-commerce to encourage demand but its connection to profitability is poorly understood because of the effects of heterogeneity of products and scale effects at the transaction level. This study compares the profit margin of a product category with discounting practices that have been observed historically based on transactional e-commerce data. Profit margins are modelled under discounting conditions using a Linear Regression model as an interpretable base and a Gradient Boosting Regressor (GBR) to extract the non-linear and interaction-based effects using some 34,500 past transactions.

The early models involving discount percentage and product category show a poor explanatory power, motivating the idea to introduce engineered features involving the absolute discount value and unit revenue as a proxy of transaction scale. R^2 , RMSE, and MAE are used to assess model performance. The findings have revealed that non-linear GBR model outperforms the linear baseline, which suggests that the sensitivity of profit to discounting is significant with respect to the scale of transactions, as opposed to the percentage or category of the product.

Sensitivity analysis of model through simulation and interpretation of the model with partial dependence plot and SHAP value indicate that historically, higher discounts are related to larger margins transactions and not causal profit impact. The results of this paper point at the weaknesses of homogenous discounting strategies and the significance of scale-conscious modelling in the study of profitability under discounting, but also the observational aspect of the past and the bias associated with selection in historical data.

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1. Introduction

1.1. Background

E-commerce has turned out to be a highly competitive market since the consumers can easily compare prices of various providers. This has led to discounting becoming a tool of establishing the level of sales, customer retention, and gross profit, as opposed to a method of marketing product sales (Lv et al., 2020). Promotions can boost demand. Nonetheless, when they are overdone or not indicated towards a certain group of consumers, the profits will be extremely slim and may result in a general lack of sustainability. Thus, e-commerce companies are grappling with the challenge of offering discounts that can drive sales and still leave a comfortable number of profits (Xie, Ma and Lui, 2018).

1.2. Problem Statement

The same traditional approaches of discounting continue to be used by many businesses such as constant price strategy to all products, the product wide discounts/discounts, and product decisions based on interpretation. Yet, such methods are not always based on the comparisons of the different products, categories, pricing schemes, etc. The consequence is that there is a possibility that certain products will get a significantly higher discount and this reduces the profit margin of the product whereas other products may not get enough discounts and they will lose out on sales.

Adoption of machine learning technology has been one of the major aspects of enhancing the application of historical transactional data to have a better comprehension of the influence that price modulations and promotions e.g. discounting can have on customers. The models evaluate the pricing trends as well as promotion success (Das et al., 2024). However, most of the previous research focused on maximizing revenue or sales volume, rather than evaluating the impact of discounting on profitability across multiple product categories (Xie, Li and Lui, 2020).

1.3. Paper Objective

This project examines how discount levels impact profit margins using historical data from e-commerce transactions. The purpose of this analysis is to identify which product categories have high enough margins to withstand substantial discounts without a large decrease in profitability and those that need conservative discounting.

The study also examines the effect of discount information for model performance. Discount effects on profit aren't likely to be linear due to different product pricing and margin structure. Therefore, for the project, a Linear Regression model will serve as an interpretable baseline against which to compare the results of a Gradient Boosting Regressor (GBR) model for capturing non-linear discounting effects. The results should provide guidance on discounting decisions at both the individual product and at the overall category level in a data driven manner.

So, this study aims to answer the following questions:

1. How do historically observed discounting practices relate to profit margins across product categories?
2. Do non-linear models outperform linear models in capturing discount-profit relationships?

2. Dataset Description

The dataset used in this research has historical transactional data of e-commerce represented by product level purchase. Each transaction represents a single purchase and includes information pertaining to the product's price, discounts given on purchases, number of units ordered, shipping cost, the category of product and final profit margin from that transaction. The size of this dataset is approximately 34,500 entries and it provides enough variation in products and discount levels needed for a thorough empirical evaluation.

The dataset contains historical data that includes the pricing and discounting strategies at the time of data collection; therefore, the data is impacted by business rules and market conditions that are in place at the time. Consequently, the dataset has limited applicability for developing discount optimization policies, but it is helpful to determine how discount factors are correlated with profit margins within an existing operational structure.

The used dataset features include:

- Product Category
- Discount Percentage
- Price
- Quantity
- Total Amount
- Shipping Cost
- Profit Margin (Target variable)

3. Exploratory Data Analysis (EDA)

Distribution of Discounts

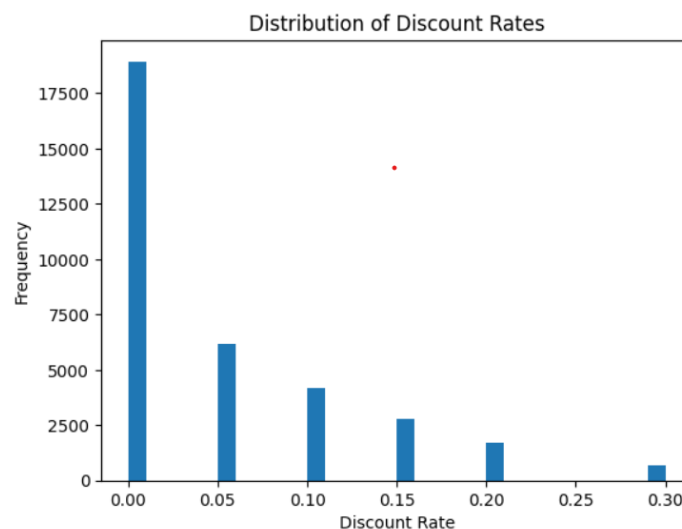


Figure 1: Distribution of Applied Discount Rates in Historical Transactions

Figure I shows the distribution of discount rates across all transactions in the dataset. Discounts are distributed unevenly, with high portions of transactions receiving no discounts, while progressively smaller numbers of transactions received large discounts.

The presence of high number of non-discounted transactions motivated the decision to filter and include only discounted transactions from the dataset. The filtering allows the analysis to focus specifically on discounting behaviours and its relationship with profit margins.

Distribution of Profit Margins

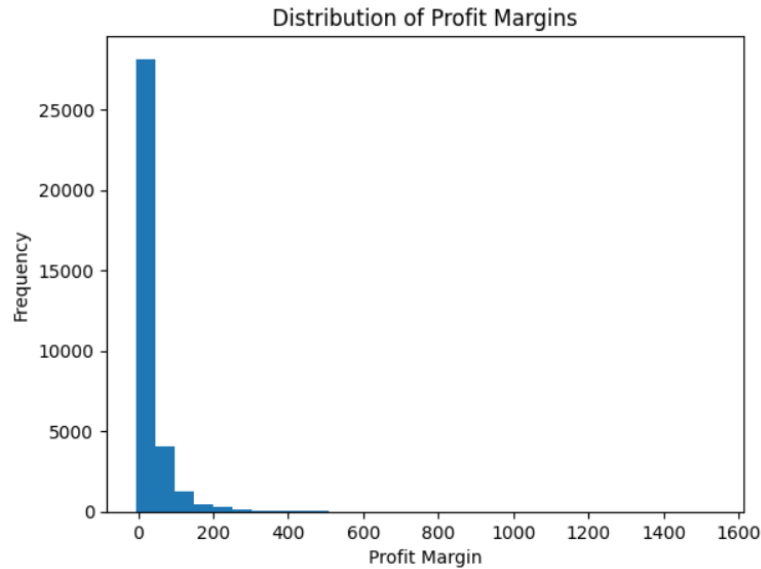


Figure 2: Distribution of Transaction-Level Profit Margins Across All Purchases

Figure II presents the distribution of profit margins across all transactions in the dataset. The distribution is highly right-skewed, with many transactions having low rates of profit, but a few with high rates. Profits vary greatly within the dataset with most of the transactions having modest profits while a small number of transactions generate unusually high profit margins.

The variation is likely caused by the differences in pricing, amount shipped and/or cost of shipping. This skewed distribution indicates the need for machine learning models capable of capturing non-linear relationships between profit margin and transaction characteristics and handling outliers more effectively.

Relationship Between Discount Rate and Profit Margin

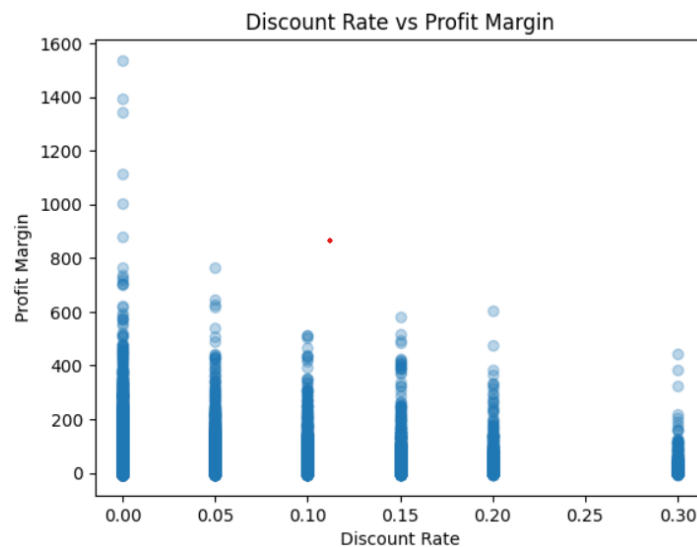


Figure 3: Relationship Between Discount Rate and Profit Margin

Figure III illustrates the relationship between discount rate and profit margin across discounted transactions. The scatter plot reveals variance in profit margin across all discount levels, especially at the lower discount rates, where low and high profit margins exist.

The absence of linear relationship is likely caused by the heterogeneity of impact of discounts on profit margins likely caused by additional factors like pricing, order quantity, shipping costs and product category. This motivates the need for a non-linear model.

Category wise Profit Margin Distribution Under Discounting

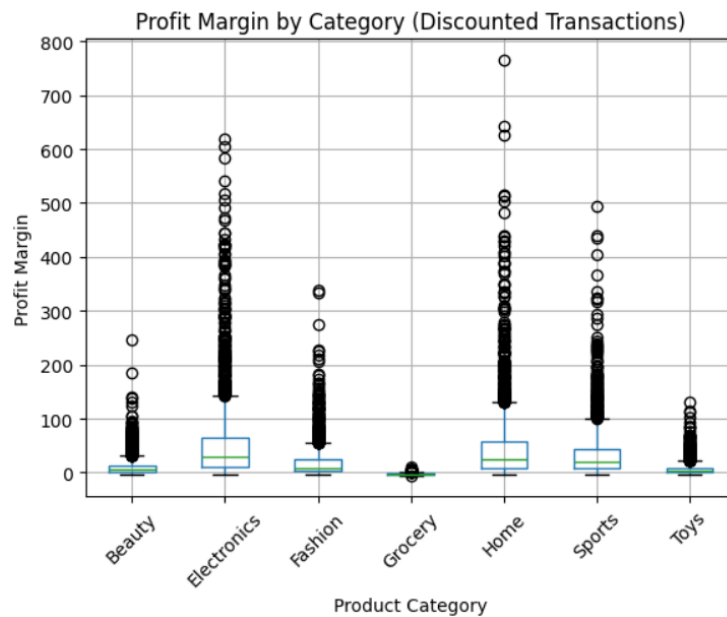


Figure 4: Category-Wise Distribution of Profit Margins Under Discounting

Figure IV presents the distribution of profit margins across product categories for discounted transactions. Significant variation is observed on the profit margins based on the product category, indicating the impact of discounting on products is dependent on category.

These observations suggest the need for a category aware modelling rather than a uniform discount strategy.

4. Feature Engineering and Data Preprocessing

Feature Engineering and Data Preprocessing played a vital role in the development of the models. The initial model was a baseline interpretation using only the available features.

4.1.Initial Feature Selection and Baseline Performance

The initial model development only utilized two main features:

- i) discount percentage
- ii) product category

The model assumed these as the primary determinants of profitability.

However, the models developed with only those features were very poor at predicting profit margins, as measured by low R^2 scores (around 0.16 and 0.17) for both linear regression and gradient boosting regressor. This indicates that these two variables are not enough to account for the variation in profit margins between transactions.

4.2.Limitations of the Baseline Feature Set

The poor performance of the model was due to the lack of sense of transactional scale in the provided features. A discount percentage doesn't concern with price, quantity or money saved. An identical discount set for a low-cost item and a high-cost item would have the same level of importance.

There was also no way to determine any cost of goods sold (COGS) metrics making the prediction unreliable.

Some features were engineered to counteract the limitations of the baseline features.

4.3.Engineered Features

Two additional features were engineered to address the limitations:

1. **Discount Value (Absolute Impact):** The discount percentage was converted into the absolute value of the discount applied.

$$\text{Discount Value} = (\text{Price} * \text{Quantity}) * \text{Discount Rate}$$

- **Impact:** This feature enables the model to quantify the exact amount reduced from an item, making it more relevant in predicting profitability than percentage of profit reduced.

2. **Unit Revenue (COGS Proxy):** This feature captures the effective price the customer paid per unit.

$$\text{Unit Revenue} = \frac{\text{Total Amount} - \text{Shipping Cost}}{\text{Quantity}}$$

- **Impact:** This feature is a strong indicator of the product's underlying cost structure and markup.

These features allow the model to understand information regarding price, size and cost structure. Allowing for a better transactional understanding.

4.4.Data Encoding, Filtering, and Splitting

The product category was encoded using one-hot encoding before the training to focus on transactions with non-zero discounts. By filtering only discounted transactions, the model could focus on profit margins under discounts rather than the decision of whether to apply a discount.

The processed data was split into training and testing subsets with an 80/20 split with fixed random seed.

5. Model Development

This study uses two supervised learning models to predict profit margins under discounts. A linear regression to act as the linear baseline model and a gradient boosting regressor for non-linear comparison. The goal is to find out if increased complexity leads to better predictive power.

5.1. Linear Regression

Linear regression was chosen as the baseline linear model to determine if profit margins and discounting relationships can be represented by linear and additive assumptions.

The model is specified as:

$$\hat{y}_i = \beta_0 + \beta_1 D_i + \beta_2 U_i + \sum_k \gamma_k C_{ik} + \varepsilon_i$$

where $\hat{y}_i$ denotes profit margin, D_i is the absolute discount value, U_i represents unit revenue, and C_{ik} are one-hot encoded product categories.

The structure allows for a transparent interpretation but assumes linearity and ignores interaction and threshold effect. Due to its simplicity the model provides a structural baseline reference and not the predictive optimum (Su, Yan and Tsai, 2019).

5.2. Gradient Boosting Regressor (GBR)

To counter the linearity of the baseline model, a gradient boosting regressor was employed. GBR constructs an ensemble of regression trees trained sequentially to minimize prediction error, given as:

$$\hat{f}(x) = \sum_{m=1}^M h_m(x), \quad h_m \in \text{Regression Trees}$$

GBR allows marginal effects to differ according to the regions of the feature space allowing for non-linear, threshold based and interaction driven relationships.

This is suitable for the discount sensitivity model as a similar discount value could have varied profit margins depending on scale and product category (Li, Li and Xu, 2018).

5.3. Training Procedure

Both models were trained using the same filters and pre-processed dataset to ensure fair comparison.

The linear regression was estimated using Ordinary Least Squares (OLS) without regularization. This approach allows high interpretability of coefficients while being straightforward baseline model (Hutcheson, 2011).

The gradient boosting regressor model used the default hyperparameter values that come with the algorithm. This method reduces the risk of overfitting due to extensive search for optimal values. This ensures the comparison between the models are based on functionality rather than optimization.

6. Performance Evaluation

6.1. Evaluation Metrics

The performance of the model was assessed by the Coefficient of Determination (R^2), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). R^2 indicates the percentage of variation in profit margin explained by the model while RMSE and MAE measure the extent of prediction error with MAE being less affected by outliers than RMSE (Kubra Serefoglu Cabuk et al., 2024).

6.2. Comparison Between Linear and Non-linear Models

Model	R^2 Score	RMSE	MAE
Linear Regression	0.6626	25.94	13.60
Gradient Boosting Regressor	0.7765	21.11	9.32

Table 1: Predictive Performance Comparison of Linear and Non-Linear Models

Table I summarizes the predictive performance of the linear regression and gradient boosting regressor models on the discounted transaction dataset.

This represents an approximate 11% point increase in explained variance and a 19% reduction in RMSE, indicating that non-linear relationships captured by the GBR model significantly improve predictive accuracy.

6.3. Linear Regression Coefficients

Feature	Coefficient
Intercept	6.415694
discount value	0.413822
unit revenue	0.155640
category Electronics	-31.928803
category Fashion	1.148545
category Grocery	-11.181027
category Home	10.921887
category Sports	7.775258
category Toys	-3.065452

Table 2: Estimated Coefficients from Linear Regression Profit Margin Model

The linear regression model provides an interpretable baseline for understanding average effects of engineered features and product categories.

Key findings include:

- **Discount value** exhibits a positive coefficient (0.41), indicating that larger absolute discounts are associated with higher profit margins on average.
- **Unit revenue** also shows a positive association (0.16), reflecting the role of transaction scale and underlying markup in determining profitability.

- Category coefficients reveal structural differences in profitability, with Electronics showing a strong negative baseline effect and Home and Sports categories exhibiting higher average margins.

While the linear model captures broad trends, its performance is limited by the assumption of linearity and additive effects. Error analysis shows increasing prediction error at higher discount levels, suggesting that discount effects on profit margins vary non-linearly across transactions.

6.4.Gradient Boosting Regressor Feature Importance

Feature	Importance
discount value	0.562595
unit revenue	0.308578
category Electronics	0.118516
category Fashion	0.000698
category Grocery	0.002442
category Home	0.004499
category Sports	0.000000
category Toys	0.000413
category Beauty	0.002259

Table 3: Relative Feature Importance in Gradient Boosting Profit Margin Model

The GBR model substantially outperforms the linear baseline by capturing non-linear interactions between pricing, discounting, and transaction scale.

Feature importance analysis indicates that:

- **Discount value** is the most influential feature (importance = 0.56).
- **Unit revenue** is the second most important predictor (importance = 0.31).
- Product category effects are present but secondary relative to transaction-level financial variables.

This confirms that profitability under discounting is driven primarily by transaction scale rather than discount percentage or category alone.

7. Model Behaviour Analysis

7.1. Partial Dependence Analysis

Partial dependence plots provide insight into the marginal effects of key predictors on profit margin.

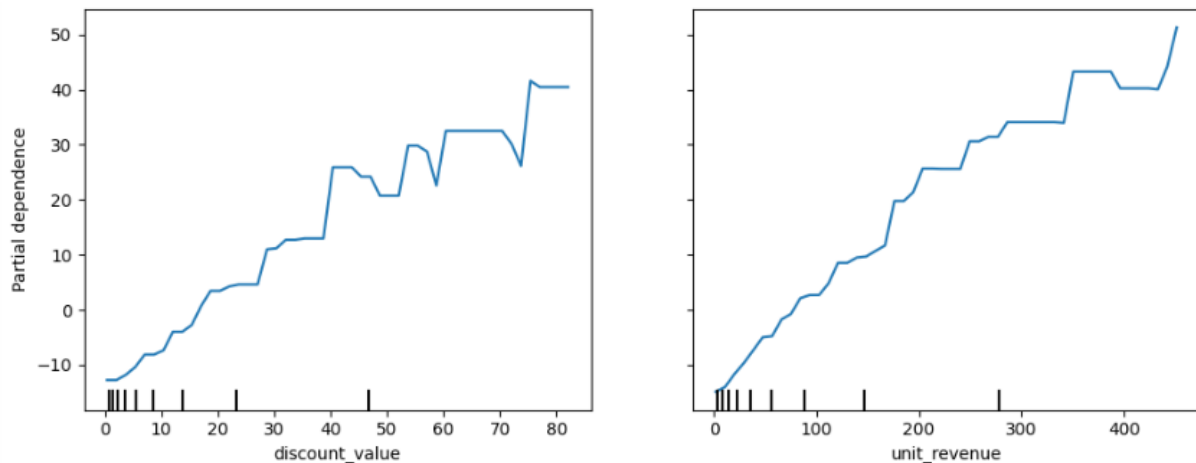


Figure 5: Partial Dependence of Profit Margin on Discount Value and Unit Revenue

Discount amount:

The relationship between profit margins and discounts is primarily non-linear but tends to be positive. Profit margins rise quickly at lower and moderate discounts and then seem to plateau at higher levels of discounting. This indicates that typically, higher discounts tend to be given toward larger ticket items where there has been enough margin for the discount to be absorbed without affecting the overall profitability of the sale.

Revenue per Unit:

Unit revenue shows a substantial and nearly monotonic trend toward profit margin; profitability will rise quickly as unit revenue increases. This increase occurs mainly in the low-to-middle ranges; as unit revenue strengthens through the upper end of the scale, this trend appears to taper off. Therefore, we can conclude that unit revenue is a reasonable indicator of product markup and product cost structure.

7.2.SHAP-Based Model Interpretation

SHAP analysis was used to explain the GBR model's predictions at both the global and interaction levels.

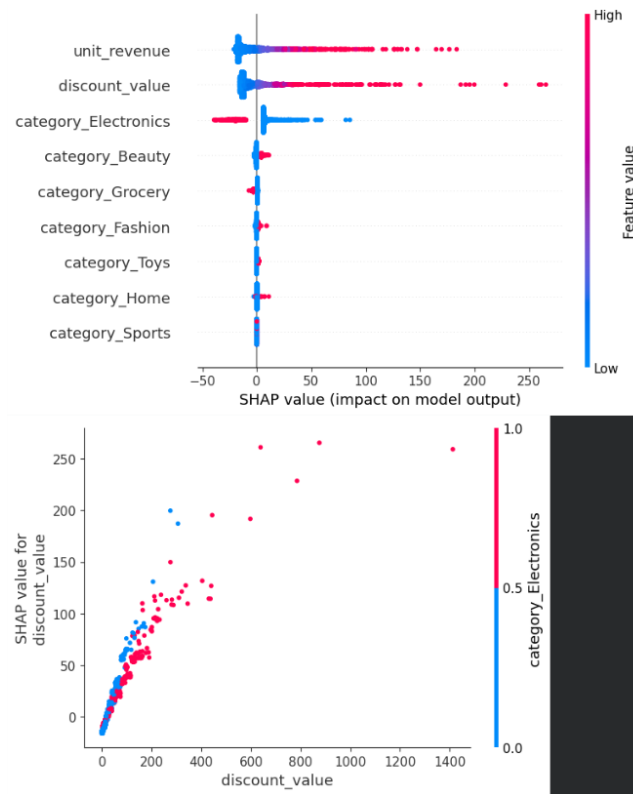


Figure 6: SHAP Summary of Feature Contributions to Profit Margin Predictions

The summary plot of SHAP clearly shows that:

- A positive effect on predicted profit margins is a result of high unit revenue.
- Predictions of profit margins are also positively affected by discount values. However, predictions involving discount value will experience more variability because of the many different discount values.
- Although the effects of category (specifically Electronics) do have some impact on profit margin prediction, category does not play as large of a role as features that are related to transaction scale.

When discount value was analysed in relation to the Electronics category, it was found that as discount value increases, Electronics transactions provide a larger positive benefit to profit margins compared to other categories. Therefore, because Electronics serves as the typical high-priced and marked up category, they can absorb large absolute discounts compared to other categories without a significantly negative effect on profit margin.

7.3.Simulation-Based Discount Sensitivity Analysis

To further understand how the models respond to changing discount levels, analysis based on simulations were performed. The primary goal of the simulations was to analyse the model behaviour in a controlled fabricated environment created from the historical data from the dataset. It mainly answers the question “given a discounted transaction, how does scale interact with discount?”.

7.3.1. Simulation Design and Assumptions

In this simulation, all factors except for discount level were held constant (that is, no other changes were made to the transaction scenario). Therefore, we were able to examine the connection between discounts and profits learned by the models.

The following assumptions were applied:

- Quantity was fixed at one unit for all simulated transactions to eliminate scale effects from order size.
- A baseline pre-discount price was derived from historical averages of unit revenue and discount value in the training data.
- For each simulated discount level, both discount value and unit revenue were updated to align with accounting logic:
 - Discount value increases linearly with the discount rate.
 - Unit revenue decreases proportionally as the discount rate increases.
- Product category was represented using one-hot encoding, with simulations performed separately for each category.

Discount rates were systematically increased between 1% and 50%, in evenly spaced steps. The models used were the same as those trained through earlier testing (linear regression and gradient boosting) and did not require further training for this research purpose.

7.3.2. Linear vs Non-Linear Discount Response

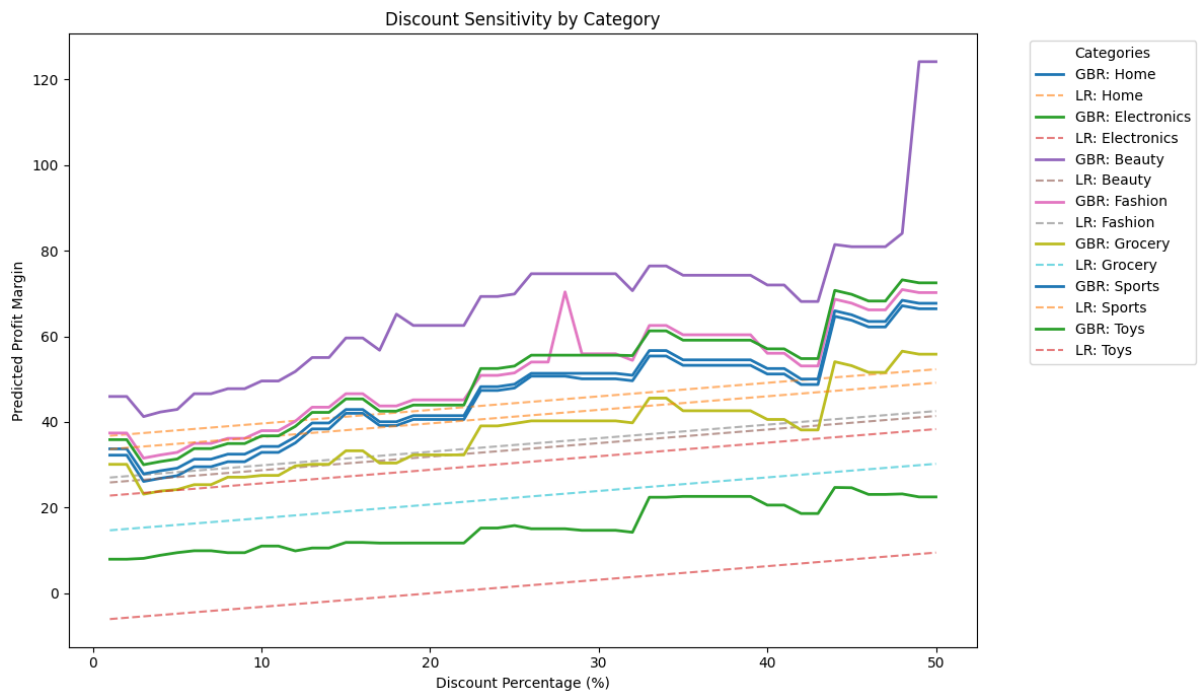


Figure 7: Simulated Profit Margin Response to Increasing Discounts: Linear vs Non-Linear Models

According to figure VII, this analysis compared relationship between discount profitability using linear regression models vs. gradient boosting regressor models for all product categories.

Linear regression model has smooth, monotonic changes as discounts increase, they cannot capture some of the more complicated relationships between profit and discounting, that's likely why it also gives low overall profit prediction.

On the other hand, GBR Models produced a piecewise or non-linear interpretation of the profit margins associated with increasing discount levels. A further observation is that within certain ranges of discount levels there are noticeable jump-like changes in predicted profit margins. The model can capture more nuanced connections between profit margins and discounts.

7.3.3. Category-Wise Discount Sensitivity

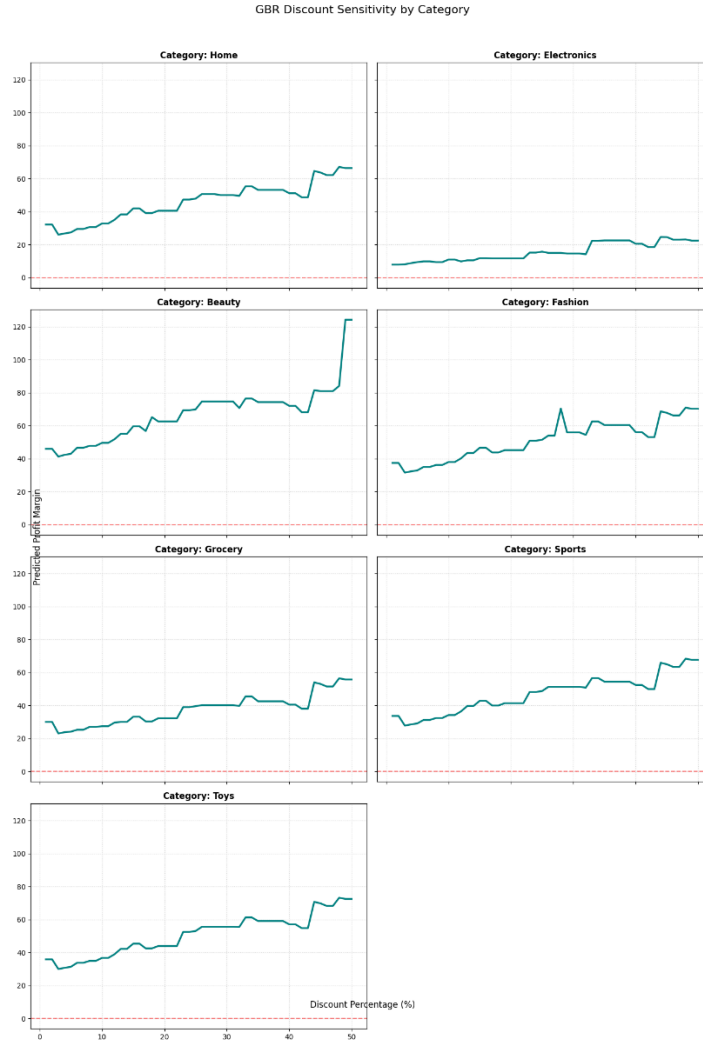


Figure 8: Category-Specific Simulated Profit Margin Sensitivity to Discount Levels

To determine whether discount sensitivity differs depending on the product category, a GBR simulation was performed independently for each category, while all other variables remained constant. The discount/profit margin curves are illustrated in Figure VIII for each of the categories tested.

As seen from the results, predicted profit margins differ between categories as all the categories have varied curves; for example, category Beauty has the highest overall profit margins while categories such as Sports, Home, Fashion and Toys have a moderately high profit margin. Category Grocery has a moderate profit margin distribution while Electronics has the lowest profit margin across all discount level.

The differences shown are due to the historical pricing structure of the categories and not directly the effect of discounting on profit margins.

As expected, the predicted profit margins show an increase with discounts in several categories. However, this trend should not be interpreted as evidence that higher discounts lead to greater profits. This reflects that high margin items historically are associated with high discounts. So, the model has learned to associate larger discount values with profitable transactions.

7.3.4. Interpretation and Limitations of the Simulation

The findings of the simulation should be interpreted as the model's learned behaviour based on the data and rules that it learned on, rather than being seen as a hypothetical economic outcome. As the models were trained on historical data that included certain discounting practices, the relationship learned naturally includes effects of selection and limitations put in place by the discounting policy.

In particular:

- The larger the discount from the dataset, the more frequently that discount is associated with higher-value transactions or a lower overall percentage of profits on that transaction.
- The simulation is therefore made based on how the model extrapolates within that learned relationship, not on how a new discount strategy, or a new way of generating revenue, would yield different results.
- As with the GBR simulation, both the non-linear and step-like patterns seen in the GBR simulation are representative of tree-like decision-boundaries, not a smooth reaction curve for the greater profitability effects of the discount change.

Therefore, while these simulations have been helpful for understanding how likely profit margin predictions will change based on any controlled discounting practice, they provide no basis for claiming that any discount level represents an 'optimal' or 'ideal' discounting strategy or that any profit increases will occur as a direct result of adopting a new discount strategy.

7.4. Residual Diagnostics

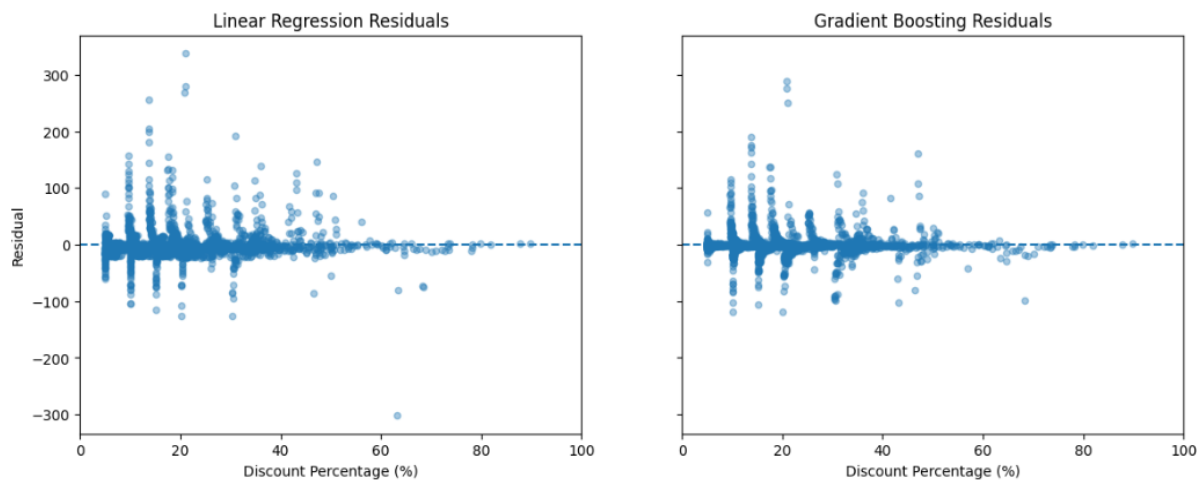


Figure 9: Residual Distribution Across Discount Levels for Linear and Gradient Boosting Models

A residual analysis of real test data was conducted to complement the simulated behaviour analysis. Residuals represent the differences between actual and predicted profit margins, and the analysis was conducted by displaying the residuals versus the discount percentage to assess the error patterns in each discount regime.

Low to mid-range discount zones contain most of the data density; therefore, they produced the maximum number of errors. Throughout all discount zones, the residual spread of the linear regression model is significantly larger and has a more irregular distribution than the residual spread of the gradient boosting model. The breadth and irregularity of the residuals suggest that the linear regression model cannot account for all the heterogeneous discount responses within one linear structure.

On the other hand, the gradient boosting model has a tightly clustered residual spread and a lower number of extreme values compared to the linear regression model. The gradient boosting model will produce a more stable prediction of profit margins than the linear regression model over the range of discount levels. While neither model provides completely noise-free predictions, the lower residual variance produced by the gradient boosting model is consistent with its ability to capture interaction effects and non-linear relationships present in the data.

7.5. Conclusion for the behavioural analysis

Key findings include:

- Increase in discount value and unit revenue shows an increase in predicted profit margins. This is because historically higher discounts are given to higher margin items.
- Features involving transaction scale play a more important role for profit margins than the category of the product itself.
- While the transaction scales do matter, the model can associate certain transactional patterns for certain categories giving a deeper understanding of the category's discounting ability.
- Residual Diagnostics shows that majority of the model error lie in low-mid discount range. While none of the model provides a clean error free predictions, the gradient boosting regressor is more tightly clustered.
- Overall, the linear model performed worse than the non-linear model in almost every aspect.

8. Conclusion, Limitations and Future Enhancements

The study examined the relationship between discount and profit margins with e-commerce transactions based on historical data using machine learning models.

The first baseline model was trained on only the provided features which did not include any information for the transactional scale. This resulted in the models having poor performance. So, some features were engineered such as the absolute discount value and the unit revenue. This significantly improved the model's performance.

Through comparative testing of model types, it was seen that Gradient Boosting Regressor (GBR) was superior to the Linear Regression model. Suggesting that linear assumptions cannot quantify the overall relationship between discounting and profitability.

The findings of this research showcase how both the models respond to different levels of discounting using sensitivity analysis via model simulation. Overall, the models were able to show the category sensitivity to discounting for historical data.

8.1. Limitations of the study

1. The data set indicates discounts are determined historically rather than at random so that it is not possible to determine the impact of the discount on the profit margin.
2. There is no explicit cost of goods information so revenue per unit is being used as a proxy for determining true product profitability.
3. The focus of the analysis on only discounted transactions means that discount adoption decisions cannot be evaluated and introduces selection bias.

4. Simulation assumptions have made simplifying assumptions regarding quantity and price and have eliminated the ability to model demand elasticity and customer behavioural responses to discounts.
5. The use of gradient boosting techniques increases predictive accuracy, however, the explanations provided by SHAP and partial dependence show the reasoning of the model rather than actual economic processes.

8.2.Future Enhancements

1. Instruments COGS:
Including COGS as part of the input data would allow direct modelling of actual profit from discount sales.
2. Include Non-Discounted Transactions:
Including both discounted and non-discounted transactions would eliminate selection bias.
3. Model Demand Response to Discounts:
Using changes in quantity elasticity or sales volume would provide a better understanding of how customer demand changes because of discounts.
4. Use Causal Inference:
Techniques including propensity score matching and causal forests could help to differentiate between policy-based correlations and the actual causal effect of discounts.

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